

Agentic AI

AI Fundamentals Series

Introduction

Agentic AI represents one of the most significant and transformative shifts in the evolution of artificial intelligence in 2025 and 2026. Unlike traditional AI systems that respond to a single prompt and produce a single output, agentic AI systems are designed to pursue goals autonomously over extended periods of time, planning sequences of actions, making decisions at each step, using tools and external resources, and adapting their approach based on feedback from the environment. The term agent in this context comes from the concept of an autonomous agent, an entity that perceives its environment, reasons about what actions to take, and executes those actions in pursuit of a defined objective, much like a human employee might approach an open-ended task. As of 2026, agentic AI has moved from research prototypes into production deployments across industries, representing a fundamental expansion of what it means for an AI system to be useful.

What Makes AI Agentic

The key distinction between a conventional AI model and an agentic AI system lies in autonomy, multi-step reasoning, and the ability to interact with the world beyond simply generating text. A conventional large language model, given a question, produces an answer and stops. An agentic system, given a goal, will determine what steps are needed to reach that goal, execute those steps in sequence, check the results of each step, and adjust its plan if something goes wrong, all without requiring human input at every stage. This loop of perceive, plan, act, and reflect is what defines agentic behavior.

To accomplish this, agentic AI systems are typically equipped with tools, which might include the ability to search the web, execute code, read and write files, send emails, query databases, call external APIs, or interact with other software applications. The underlying large language model acts as the reasoning engine, deciding which tool to use, when to use it, and how to interpret the results, while the tools extend the system's reach beyond pure language into the real world of data and action.

Planning and Memory

Two capabilities that are particularly important for agentic AI are planning and memory. Planning refers to the system's ability to break down a high-level goal into a series of subtasks and determine the order in which those subtasks should be completed. This might involve creating an explicit plan at the start of a task, or it might involve more dynamic, reactive planning where the system decides its next step based on the outcome of the previous one. Techniques such as chain-of-thought reasoning, where a model thinks through a problem step by step before acting, and tree-of-thought reasoning, where multiple possible approaches are considered and the most promising is selected, are commonly used to enhance planning capability.

Memory is equally important, since agentic tasks can span many steps and often require the system to retain information gathered or generated earlier in the process. Agentic systems typically maintain a working memory of the current task's state and history, which is passed along with each prompt to the underlying model. Some systems also use external memory stores, such as databases or vector stores, to retain information across sessions, enabling a form of long-term memory that allows the agent to accumulate knowledge and context over time.

Multi-Agent Systems

One of the most active areas of development in agentic AI involves multi-agent systems, in which multiple specialized AI agents work together to accomplish complex tasks, each handling the parts of a problem it is best suited to address. In such a system, an orchestrator agent might receive a high-level goal and delegate subtasks to specialist agents, such as a research agent, a coding agent, a writing agent, and a quality-checking agent, coordinating their outputs into a final result. This division of labor mirrors the way human organizations work, with different team members contributing their expertise to a shared project.

Multi-agent architectures introduce their own challenges around coordination, communication, conflict resolution, and ensuring that the overall system behaves coherently even when individual agents operate with significant autonomy. Protocols for agent-to-agent communication, such as Anthropic's Model Context Protocol and Google's Agent-to-Agent protocol, have emerged to standardize how agents share information and delegate tasks, making it easier to build robust multi-agent pipelines.

Real-World Applications

Agentic AI is already delivering tangible value across a wide range of domains. In software development, agentic coding assistants can take a natural language description of a feature, write the code, run the tests, identify failures, debug the issues, and iterate until the tests pass, dramatically accelerating the development process. In customer service, agentic systems can handle complex, multi-step support interactions, looking up account information, processing requests, coordinating with other systems, and escalating to a human only when genuinely necessary. In research and analysis, agentic AI can conduct literature reviews, gather data from

multiple sources, synthesize findings, and produce structured reports on topics that would take a human researcher hours or days to compile.

In business operations, companies are deploying agentic workflows to automate supply chain monitoring, financial reconciliation, compliance checking, and market research. Scientific research organizations are using agentic AI to design experiments, analyze results, and generate hypotheses, with some systems running autonomous experimental loops in which the agent iterates through cycles of hypothesis, experiment, and analysis with minimal human oversight.

Challenges and Risks

Despite its promise, agentic AI introduces serious challenges that must be addressed carefully. The greater autonomy of these systems means that errors can propagate and compound across many steps before a human has the opportunity to intervene, potentially leading to significant unintended consequences, particularly in high-stakes environments. Ensuring that an agentic system's goals remain aligned with what its operators actually want, a problem sometimes called goal specification or alignment, is considerably harder when the system is making many sequential decisions rather than answering a single question.

Safety mechanisms such as human-in-the-loop approval for high-risk actions, sandboxed execution environments that limit what a system can actually affect, and robust monitoring and logging of agent behavior are considered essential best practices for responsible agentic AI deployment. Researchers and practitioners also emphasize the importance of setting clear boundaries on what an agent is permitted to do, since the flexibility that makes agentic systems powerful also makes them capable of taking surprising and potentially harmful actions if those boundaries are not carefully defined.

Conclusion

Agentic AI marks a genuine inflection point in the practical utility of artificial intelligence, moving from systems that answer questions to systems that pursue goals, take actions, and deliver results across extended, complex workflows. Its combination of reasoning, tool use, planning, and multi-agent coordination is reshaping what organizations can accomplish with AI, and the pace of adoption in enterprise settings suggests that agentic systems will become a foundational layer of business operations in the years ahead. The central challenge for the field is ensuring that the autonomy these systems possess is matched by appropriate mechanisms for oversight, alignment, and accountability.